



Adsorption processes for environmental sustainability

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This special issue of Environmental Science and Pollution Research (ESPR) presents selected contributions from the 15th Brazilian Meeting on Adsorption (EBA 15), held in Maceió, Alagoas, Brazil, from November 20 to 22, 2024. Researchers used the meeting to present recent advances, debate current challenges, and sketch where adsorption science and related fields may go next. Professor Lucas Meili of the Federal University of Alagoas, Brazil, organized EBA 15.

Since its first edition, the Brazilian Meeting on Adsorption (EBA) has grown into one of Brazil's main scientific forums for adsorption science. It is held biennially, in a different region of the country each time. Researchers, students, professionals, and industry representatives come to exchange ideas, discuss emerging challenges, and share advances in fundamental and applied adsorption. The meeting has also done something harder to measure: it strengthened the Brazilian adsorption community, brought research groups into contact with one another, and widened the community's links with colleagues abroad.

EBA has always mixed research, education, and scientific networking. The clearest example is the Adsorption School, where students and early-career researchers work through fundamental concepts, experimental approaches, and emerging applications in lectures given by experienced

researchers. The meeting communicates new results and trains the people who will produce the next ones.

The 15th edition reflected this broad scope. EBA 15 gathered 198 participants and 193 scientific contributions, presented in oral and poster sessions. The program included 16 invited speakers and researchers from institutions in Brazil, Portugal, Argentina, Colombia, the United States, and the United Kingdom, reflecting the meeting's growing international reach. The Adsorption School comprised five specialized lectures and remained a core part of the scientific and educational program.

The contributions presented at EBA 15 show how diverse the field has become. Topics ran from the development and characterization of new adsorbent materials to environmental remediation, CO₂ capture and storage, biotechnology, catalysis, energy applications, and the modeling and simulation of adsorption processes. The meeting also covered advanced porous materials, hybrid adsorbents, and work aimed at moving adsorption technologies out of the laboratory and into practical and industrial use.

Selected contributions from the meeting were invited for submission to this special issue of Environmental Science and Pollution Research. Seven research articles were accepted after the journal's regular peer-review process, reflecting the quality of the work and the conference's thematic range.

The accepted articles cover a broad range of adsorption topics, with emphasis on environmental applications and advanced adsorbent materials. They address modified clays for dye removal, porous carbons derived from waste materials for CO₂ capture, magnetic and conducting polymer-based composites for adsorbing textile dyes, and nanostructured materials for treating oilfield-produced water. Other studies explore Ca-modified biochar for phosphate removal, zeolite-geopolymer composites for adsorbing toxic metals, and biochars produced from agro-industrial residues for removing methylene blue.

Together, these studies point to the directions now shaping adsorption science: valorization of waste and biomass-derived materials, the design of multifunctional and nanostructured adsorbents, the removal of emerging and conventional contaminants from aqueous systems, and adsorption-based strategies for carbon capture and environmental

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remediation. The collection is interdisciplinary by nature. It draws on materials science, environmental engineering, process development, and adsorption fundamentals.

Authors contributions Conceptualization: [G. L. Dotto, M. A. da Motta Sobrinho, L. Meili]; Methodology: [G. L. Dotto, M. A. da Motta Sobrinho, L. Meili]; Formal analysis and investigation: [G. L. Dotto, M. A. da Motta Sobrinho, L. Meili]; Writing—original draft preparation: [G. L. Dotto, M. A. da Motta Sobrinho, L. Meili]; Writing—review and editing: [G. L. Dotto, M. A. da Motta Sobrinho, L. Meili]; Funding acquisition: [G. L. Dotto, M. A. da Motta Sobrinho, L. Meili]; Supervision: [G. L. Dotto, Maurício A. da Motta Sobrinho, L. Meili]. All authors read and approved the final manuscript.

Funding The authors thank CAPES (Coordination for the Improvement of Higher Education Personnel), CNPq (National Council for Scientific and Technological Development), and FAPEAL (Alagoas State Research Foundation) for the financial support.

Data availability The datasets used and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent to publish Not applicable.

Competing interests Prof. Guilherme Luiz Dotto declares that he is Editor of Environmental Science and Pollution Research. The other authors declare no competing interests.

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